

3 (a) Explain the difference between *accuracy* and *precision* and state the type of error associated with each.

(i) Accuracy:

Accuracy refers to how closely a measured value agrees with the true value.

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Type of error: systematic

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(ii) Precision:

Precision refers to how closely individual measurements agree with each other (or average value) without reference to any true value.

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Type of error: random.....[2]

(b) Three digital clocks A, B and C are being tested in a laboratory. Using signals from the Global Positioning System, the displays on the clocks at the exact time of 12:00:00 on four successive days are shown in **Fig. 1.1**. The clocks are reset each day at 00:00:00.

Clock	Day 1	Day 2	Day 3	Day 4
A	12:06:40	12:06:38	12:06:39	12:06:43
B	12:03:59	12:02:49	12:01:54	12:03:15
C	11:59:59	12:00:02	11:59:57	12:00:05

Fig. 1.1

State and explain,

(i) which clock is the most accurate, and

Clock C is the most accurate. The average is closest to the true value.

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.....[1]

(ii) which clock is the most precise.

Clock A is the most precise. The variance is the least.

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.....[1]

(c) For clock B in **(b)**, estimate the time displayed at noon on day 5 with its associated uncertainty.

Mean: $12\text{h} \left[(180 + 59) + (120 + 49) + (60 + 54) + (180 + 15) \right] / 4 \text{ seconds} = 12\text{h} \ 179\text{s}$

Time: 12:02:59

Range/2 = 63 $\approx$ 70 s OR Mean – Min = 65 $\approx$ 70 s

Time: 12:02:59 $\pm$ 70 s

Time = $\pm$ s [2]

4 **(a)** Explain what is meant by an *ideal gas*